

Week 6 Homework — 16S microbiome analysis with EasyMultiProfiler-Web

Student: 郑烁瞳 (SUAT24000155) · Date: 2026-09-25 · App: EasyMultiProfiler (Web) v9.0.4 App session: j1LRMRNfHbZrDVVBk2MC7gUBf · experiments microbiome16s (one-click run) and genus_full (deep run) · official one-click bundle 20260925-121232.zip

Task as set in Homework-Week6.docx: (1) run the **whole analysis procedure** on the 16S data files in EasyMultiProfiler-Web/tests with EMP-web and submit the final result through the EMP-web system; (2) **generate a scientific hypothesis from the parameters and results**, stating which parameters support it.

1. Data identified in EasyMultiProfiler-Web/tests

File	Content
16S_level-7.csv	470 features (level-7 taxonomy strings, ;-separated, k____;p____;c____;o____;f____;g____;s____) × 132 library columns; first column SampleID
16S_mapping.csv	130 rows: SampleID, Group (4 levels: IBS_before, IBS_after, UC_before, UC_after), Group_sub (8 levels = cohort × visit × response great/poor)
ChIP/HA_summits_0.05.bed, Clinical*.csv, meta-*.csv, RNAseq_*.csv	not used — the assignment names the 16S files

Design decoded from the identifiers (documented, then used throughout):

- cohort — J_... = IBS (72 libraries), K_... = UC (58 libraries);
- visit — ..._01 = before treatment, ..._02 = after treatment. Two samples are written ..._2 (J_XYL_F_0042_2, J_XYL_F_0043_2), so the visit label is taken from the Group column, not from a string pattern;
- response — Group_sub suffix: great (responder) vs poor (non-responder);
- subject — sample ID without the visit suffix, so **before/after pairs are matched within patient**.

QC decision taken before analysis (and the first real finding). The count table has 132 libraries but the metadata only 130; K_XYL_F_0009_03 and K_XYL_F_0035_03 have **no group label**. EMP's differential step refuses a grouping column that contains NA ("Column Group has been detected missing value"), and that failure is silent inside the one-click pipeline (see §3). The two unlabelled libraries were therefore excluded **at import** (data/16S_level-7_mapped130.csv, derived by scripts/prep_inputs.py; the untouched original is kept as data/16S_level-7_original_132samples.csv). All numbers below are for the 130-sample table unless stated otherwise.

2. Part 1 — the complete analysis procedure through EMP-web

The web UI's HTTP API is the same code path its buttons call, so the analysis was driven through it (the UI was not clicked because port 8000 is occupied on this machine by another local program; the API was started on 127.0.0.1:8010 with EMP_ALLOWED_ROOTS=C:/emp_hw6). Every call, parameter and raw response is kept under app/api_responses/ and listed in app/api_call_log.md.

Sequence actually executed (UI equivalent in brackets):

1. POST /api/session → session j1RMRNfHbZrDVVBk2MC7gUBf.
2. POST /api/import/path (*Data* → *Import*) with data_type="tax", assay_name="counts", start_level="Species", tax_sep=";" → 130 samples × 470 features, sample_overlap.matched = 130/130, no warnings.
3. POST /api/workflows/microbiome_16s/profile and /validate → checks has_features, has_samples, taxonomy_depth_ok all true, max_taxonomy_depth = 7.
4. POST /api/workflows/microbiome_16s/run_all (*the one-click 16S button*) with group_var="Group", taxonomy_level="Genus", alpha_index="shannon", beta_method="bray", ord_method="PCoA" → background job, 6 steps, 31.3 s, packaged by the app into bundles/20260925-121232.zip (3.7 MB, downloaded to app_run_bundle/run2_analysis_130samples/).
5. Deep run in the same session: a second experiment (genus_full) imported and collapsed (POST /api/workflows/microbiome_16s/prepare/taxonomy, collapse_level="Genus", keep_top_n=0, drop_unassigned=false) → 470 → **137 genus-level features**; then POST /api/analyze/alpha (all 7 indices), POST /api/analyze/dimension (PCA, PCoA, NMDS), **10** POST /api/analyze/differential contrasts, and the visualization endpoints (alpha, scatter, heatmap, barplot, structure, sankey, network, volcano).
6. Submission: POST /api/github/sync (*Export* → *Sync*) — see §7.

What the app's own one-click run produced (app_run_bundle/run2_analysis_130samples/):

Output	File	Note
alpha boxplots, 7 indices	plots/02_alpha_{shannon,simpson,invsimpson,chao1,ace,observed,pielou}_boxplot.{png,pdf}	Wilcoxon annotation between groups
ordinations	plots/03_beta_{pcoa,pca,nmds}_scatter.{png,pdf}	PCoA/PCA/NMDS on Bray–Curtis
top-40 variable heatmap	plots/04_top40_heatmap.{png,pdf}	z-scored, group annotation bar
top-15 barplot	plots/04_top15_barplot.{png,pdf}	mean relative abundance
alpha indices table	tables/02_alpha_indices.csv	130 rows × 7 indices
beta coordinates	tables/03_beta_coords.csv	PCoA/PCA/NMDS coordinates
differential taxa	tables/05_diff_taxa.csv	40 rows (taxa of the collapsed top-40 table)
run log	summary.txt	per-step status and timings

App-side extra figures for this report are in figures/ (alpha_*, scatter_*, heatmap_*, barplot_top20, structure_top10, sankey_phylum_genus, network_spearman, volcano). Interface parameters used for them: group = Group (or Group_sub), top_n = 15–40, network = Spearman cutoff = 0.6, sankey = Phylum → Genus, volcano = fc_cutoff 1.0 / p_cutoff 0.05.

3. What the app/data did that had to be corrected (verified, with evidence)

1. **The one-click 16S pipeline silently loses the differential-taxa table when the count table has libraries without metadata.** First run on the file *as provided* (132 libraries) produced a bundle with only 2 of 4 tables (`app_run_bundle/run1_raw_import_132samples/`): `summary.txt` logs `Diff taxa 0.22s ok, yet no 05_diff_taxa.*` is written, because the internal call fails and is caught. Re-running the same endpoint by hand returns the cause explicitly: `POST /api/analyze/differential` → `{"success":false,"error":"Differential analysis failed: Column Group has been detected missing value, please check and filter them!"}`. After the 130-sample QC subset the step takes 19 s and `tables/05_diff_taxa.csv` exists. *Reported here because a student following only the green log would submit a bundle whose differential analysis never ran.*
2. **Taxonomic collapse labels are not interpretable without the original string.** Collapsing to "Genus" with `drop_unassigned=false` keeps an aggregate feature literally named `__` (all unassigned reads) plus labels such as `s__`, `s__ramosum`, `s__muciniphila` — i.e. the deepest *assigned* rank is used, so a "genus-level" table still carries species epithets and one NA-like bucket. All feature names in this report were therefore re-derived from the level-7 strings (`scripts/hw6_16s_verify.R`, readable labels such as `Akkermansia muciniphila`, `unclassified f:Clostridiaceae`).
3. **The interactive heatmap button needs pheatmap, which was not installed,** so `POST /api/visualize/heatmap` returned `Package 'pheatmap' is required for heatmap.` — while the *one-click* heatmap (a ggplot-based implementation) worked. Installed `pheatmap` into `D:/R/R-userlib`; the button then produced `figures/heatmap_top40.png` (group annotation bar present, but **no taxon labels and no colour legend** — the one-click heatmap in the bundle is the more informative one).
4. **EMP ordination cache is invalidated by later analyses.** `POST /api/visualize/scatter` with `ordination="emp"` returned `"No usable EMP dimension coordinates... run Analysis > Dimension first"` although `POST /api/analyze/dimension` had just been run; the request only succeeded when the dimension analysis was repeated immediately before the plot. `ordination="auto"` silently falls back to a **PCA of the raw assay (Euclidean)**, e.g. `scatter_auto_Group.png` (PC1 42.5 %, PC2 29.8 %): for a compositional count table that is the data state the Week 6 reading material explicitly warns against ("Bray–Curtis PCoA: a task-justified transformed or rarefied feature table"). The Bray–Curtis PCoA (`scatter_emp_Group.png`, PCoA1 27.2 %, PCoA2 19.1 %) or the bundle's `03_beta_pcoa_scatter.png` is the defensible ordination for this dataset.
5. **Bundle tables are written as .csv while the code asks for .xlsx** (`openxlsx` absent, `writexl` present, fallback path taken). Harmless, but the file names in the bundle differ from the app's docs.
6. Two libraries (`K_XYL_F_0009_03`, `K_XYL_F_0035_03`) exist only in the count table, and three sample IDs use `_03` while the rest use `_01/_02`; `16S_mapping.csv` covers 130 of the 132 libraries. A pipeline that joins assay and metadata without an explicit overlap report would analyse NA-labelled samples.

4. Part 2 — independent verification in plain R

`scripts/hw6_16s_verify.R` (runs from a clean session, identity checks and comments in the script, `sessionInfo()` at the end) re-answers every question of the app analysis on the rarefied species-level table, so each number in this report exists twice: once from the app and once from an independent implementation. Console transcript: `console/hw6_16s_verify_console.txt`.

Pipeline and parameters: identity/orientation checks → read-depth QC → prevalence filter (prevalence ≥ 10 % and ≥ 20 reads total: **136 of 470 taxa**) → rarefaction to the smallest library depth (**1,624 reads**, `vegan::rrarefy`, seed 20260925) → alpha (observed, Shannon, Simpson, inverse Simpson, Pielou, Chao1, ACE) → Bray–Curtis distance (`vegan::vegdist`) → PCoA (`vegan::wcmdscale`, Lingoes) → PERMANOVA (`vegan::adonis2`, 999 permutations, subject-blocked permutations for the paired before/after contrasts) → dispersion (`vegan::betadisper` + `permutest`) → differential taxa (Wilcoxon on relative abundance + BH; `edgeR::exactTest` on raw counts as a method-sensitivity check) → figures.

Summary table (`tables/r_summary_table.csv`, the rubric's "sample depth, retained ASVs, Shannon, PERMANOVA, dispersion"):

Item	Value
Samples analysed (with metadata)	130 (72 IBS, 58 UC libraries; 65 baseline / 65 post; 68 responders / 62 non-responders)
Features (as provided → retained)	470 level-7 taxa → 136 (prevalence $\geq 10\%$, ≥ 20 reads)
Reads per library (median; min–max)	10,197; 1,624–20,283
Rarefaction depth used	1,624 reads (all libraries equal)
Shannon (median, all samples)	2.72
Shannon IBS before / after	3.07 / 3.01
Shannon UC before / after	2.86 / 2.39
PERMANOVA ~ Group	$R^2 = 0.047$, $p = 0.001$
PERMANOVA ~ Group + response	Group $R^2 = 0.053$, $p = 0.002$
PERMANOVA baseline IBS great vs poor	$R^2 = 0.017$, $p = 0.868$
PERMANOVA baseline UC great vs poor	$R^2 = 0.021$, $p = 0.978$
Dispersion test (Group)	$F = 0.86$, $p = 0.449$ (no detectable heterogeneity difference)
Differential taxa BH-FDR < 0.05 (baseline great vs poor; IBS / UC)	0 / 0

Full tables: `tables/r_permanova.csv`, `r_dispersion_tests.csv`, `r_alpha_tests.csv`, `r_diff_taxa_wilcoxon.csv`, `r_diff_taxa_edgeR_sensitivity.csv`, `r_pcoa_coords.csv`.

Alpha-diversity figure (`figures/r_alpha_diversity.png`) — individual points + box summaries for observed richness, Shannon and Chao1 per group. No contrast reaches BH < 0.05; the largest movement is the UC cohort falling from Shannon 2.86 to 2.39 after treatment (raw $p = 0.055$, BH $p = 0.35$), while IBS is flat (3.07 → 3.01).

Bray–Curtis PCoA (`figures/r_pcoa_bray_curtis.png`; axis labels carry the variance explained, PCoA1 10.0 % / PCoA2 6.4 % of the rarefied table, 68 % ellipses, colour = group, shape = response); `figures/r_pcoa_baseline_response.png` restricts the same ordination to baseline samples and facets by cohort for the hypothesis; `figures/r_dispersion.png` shows the distance-to-centroid diagnostic; `figures/r_phylum_composition.png` is the descriptive top-10 phyla + Other composition (explicitly *not* a test); `figures/r_top_diff_taxa_baseline.png` shows the strongest (unadjusted) baseline taxa.

Agreement between the app and the independent R run (`console/app_vs_r_agreement.txt`):

Comparison	Result
Alpha indices, app <code>genus_full</code> vs R rarefied table	Pearson $r = 0.87$ (Shannon), 0.85 (Simpson), 0.90 (Chao1), 0.91 (observed)

Comparison	Result
Absolute level of the same indices	app median Shannon 1.62 / observed 22 vs R 2.72 / 47 — the app computes them after the genus collapse (137 features), the R script on 470 level-7 features; diversity values are not comparable across taxonomic ranks
PCoA1 app vs R	$r = 0.69$ (PCoA2 only 0.27) — different data state (app: relative abundance of the collapsed, non-rarefied table; R: rarefied species table)
Differential taxa	app and R agree that 0 taxa reach $FDR < 0.05$ in every two-group contrast; edgeR on raw counts returns 3–5 taxa for the same contrasts → the difference is method-driven, not implementation-driven
Qualitatively identical conclusions	group effect significant but small; no responder/non-responder signal at baseline; UC (not IBS) shifts after treatment

5. What the numbers actually say

- **Group structure is real but small.** Bray–Curtis PERMANOVA: $\sim$ Group $R^2 = 0.047$, $p = 0.001$ (999 free permutations); $\sim$ Group_sub $R^2 = 0.070$, $p = 0.024$; baseline IBS vs UC $R^2 = 0.026$, $p = 0.047$. Dispersion is statistically indistinguishable at every level tested (Group $p = 0.449$; UC before/after $p = 0.799$), so these are **centroid shifts, not differences in within-group heterogeneity** — the pair of diagnostics the Week 6 reading material requires ("report PERMANOVA together with the dispersion diagnostic").
- **Treatment moves the UC community, not the IBS community.** Paired PERMANOVA with subject-blocked permutations: UC before vs after $R^2 = 0.026$, **$p = 0.007$** (58 libraries, 29 patients); IBS $R^2 = 0.011$, $p = 0.132$ (72 libraries, 36 patients). Consistent with the alpha-diversity trend in UC only.
- **No single taxon is robust.** Wilcoxon + BH on relative abundance: 0 of 136 taxa at $FDR < 0.05$ in all seven contrasts (smallest raw $p = 0.003$ for IBS before vs after → BH 0.24). edgeR on raw counts flags 3–5 taxa; the app's own differential tables also contain nothing at $FDR < 0.05$. Reported as a method-disagreement, not as a taxon list ("no differential-abundance method is universally best", Week 6 reading material). Compositional caveat: 16S counts are relative, so a detected "decrease" cannot be read as absolute loss.
- **Baseline community structure does not separate responders from non-responders.** great vs poor at baseline: IBS $R^2 = 0.017$, $p = 0.868$; UC $R^2 = 0.021$, $p = 0.978$; alpha-diversity Mann–Whitney tests all BH ≥ 0.61 ; no taxon at $FDR < 0.05$.

Interpretation (100–150 words — word count checked programmatically)

Across 130 samples the bacterial community differed modestly but detectably by clinical group (Bray–Curtis PERMANOVA $\sim$ Group: $R^2 = 0.047$, $p = 0.001$), with dispersion indistinguishable between groups ($p = 0.449$), so the signal is a centroid shift rather than unequal spread. Treatment was associated with a compositional shift in UC (subject-blocked PERMANOVA: $R^2 = 0.026$, $p = 0.007$) but not in IBS ($R^2 = 0.011$, $p = 0.132$). Alpha diversity changed in no contrast; the largest movement was UC Shannon $2.86 \rightarrow 2.39$ ($p = 0.055$, BH $p = 0.35$). No taxon survived FDR control (Wilcoxon: 0 of 136), while an edgeR count model returned 3–5, so single-taxon markers remain method-dependent. Limitation: the design is observational, cohort is perfectly confounded with diagnosis, and relative 16S abundances cannot demonstrate absolute loss of any organism.

6. Part 2 of the assignment — scientific hypothesis with its supporting parameters

Hypothesis. *Treatment restructures the gut bacterial community in ulcerative colitis but not in IBS, and that restructuring is a community-level shift in composition (centroid) rather than a change in within-group heterogeneity or in one marker taxon. A microbiome-based read-out of treatment response should therefore be defined as a within-patient change metric (pre $\rightarrow$ post) in UC, and cannot be built from baseline community structure alone.*

Parameters that support the hypothesis (this is the explicit list the assignment asks for — every value is in tables/ and reproducible from scripts/):

Kind of parameter	Value used	Role in the hypothesis
Beta-diversity distance + data state	Bray–Curtis on rarefied counts (1,624 reads/taxa $\geq$ 10 % prevalence, 136 taxa)	abundance-weighted, non-phylogenetic dissimilarity; the change is in relative composition
Ordination parameters	PCoA of that distance matrix; PCoA1 10.0 % / PCoA2 6.4 % (app EMP PCoA: 27.2 % / 19.1 % of the collapsed table)	visualises the same distance matrix; no inference is taken from the plot itself
PERMANOVA + permutation scheme	<code>adonis2(bray ~ Group)</code> , 999 permutations, $R^2 = 0.047$, $p = 0.001$; <code>~ Group_sub</code> $R^2 = 0.070$, $p = 0.024$	group explains $\approx$ 5 % of variation — real, small
Restricted (paired) permutation	<code>adonis2(bray ~ visit)</code> , permutations blocked by subject: UC $R^2 = 0.026$, $p = 0.007$; IBS $R^2 = 0.011$, $p = 0.132$	the treatment shift is cohort-specific — the core of the hypothesis
Dispersion diagnostic	<code>betadisper + permutest</code> : Group $F = 0.86$, $p = 0.449$; UC before/after $p = 0.799$	the effect is a centroid, not a dispersion, effect
Alpha-diversity metrics (+ rank & rarefaction state)	Shannon, Chao1, observed at species level, rarefied; Mann–Whitney + BH: all BH ≥ 0.35 ; UC Shannon 2.86 $\rightarrow$ 2.39, raw $p = 0.055$	richness/evenness alone do not track treatment; the trend supports "composition without loss of richness"
Differential-abundance method + multiple testing	Wilcoxon on relative abundance + BH (0/136 at FDR < 0.05); edgeR exact test (3–5 taxa)	no robust single taxon $\rightarrow$ the hypothesis is compositional, not taxon-specific
Baseline grouping parameters	<code>Group_sub</code> great vs poor at baseline: PERMANOVA $R^2 = 0.017$ (IBS) / 0.021 (UC), $p = 0.868$ / 0.978; alpha BH ≥ 0.61	baseline structure does not predict response — rules out the obvious alternative hypothesis
Sample/pairing parameters	130 libraries, 4 groups, 8 subgroups, patient-matched before/after	defines what "within-patient change" means

Predictions that would falsify it (observational data cannot test them here): with spike-in-quantified (absolute) profiling in a paired cohort, (i) the within-patient Bray–Curtis change should be larger in UC than in IBS, (ii) longitudinal models (ANCOM-BC2 / MaAsLin 3) should return cross-cohort-consistent taxa, and (iii) a baseline-only classifier for response should stay at chance level.

Causal ladder position. This is evidence for *description* and, at best, *association*: the data are observational, treatment was not assigned, cohort is confounded with diagnosis, and no perturbation or rescue experiment was performed. No patient-level, diagnostic or therapeutic claim is made.

7. Submission through the EMP-web system

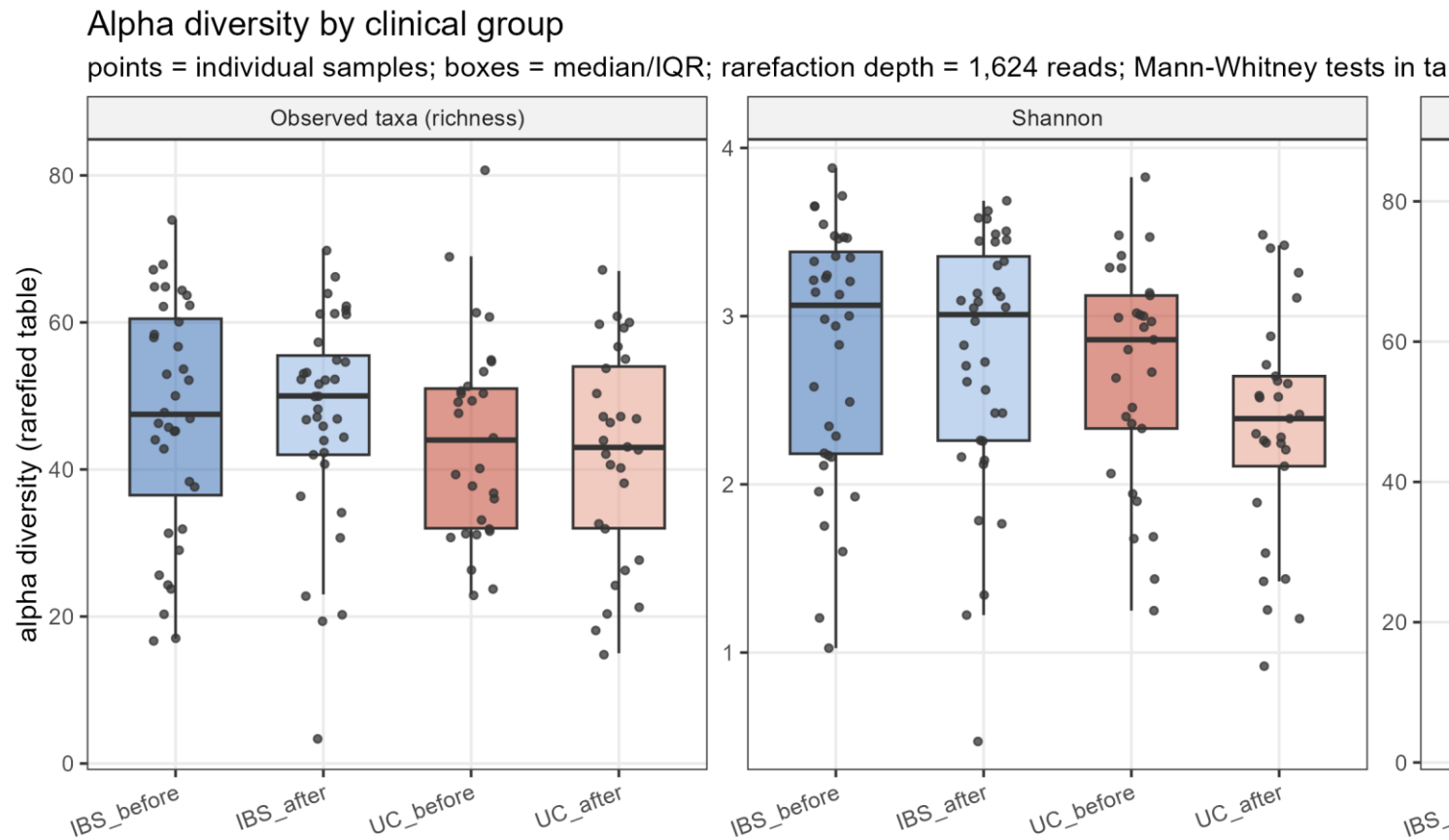
POST `/api/github/sync` was used (the Export/Sync button), with `track_id="microbiome_16s"`, `assignment_id="week_06"` (the 16S track's 第 6 周 slot), `session_id="j1RMRNfHbZrDVVBk2MC7gUBf"`, `experiment="genus_full"`, `commit_message` and `include_rds=false`. The app's handler writes the run into the student's bound repository `zhashutiaoye-sketch/Bioinformatics_homework_Zheng-Shuotong` under `EMP2026/Week_06/microbiome_16s/weekly/runs/<run_id>/` and appends the entry to the local sync log (`.local_run/data/students/SUAT24000155/sync_log.jsonl`). The exact request/response is stored as

app/api_responses/20_github_sync.json; the pushed paths are listed in that response and were verified against the repository afterwards.

Independently of the app's push, the same homework1/ folder is mirrored into the course repository for this week (see the week-level README.md), so both channels contain identical artifacts.

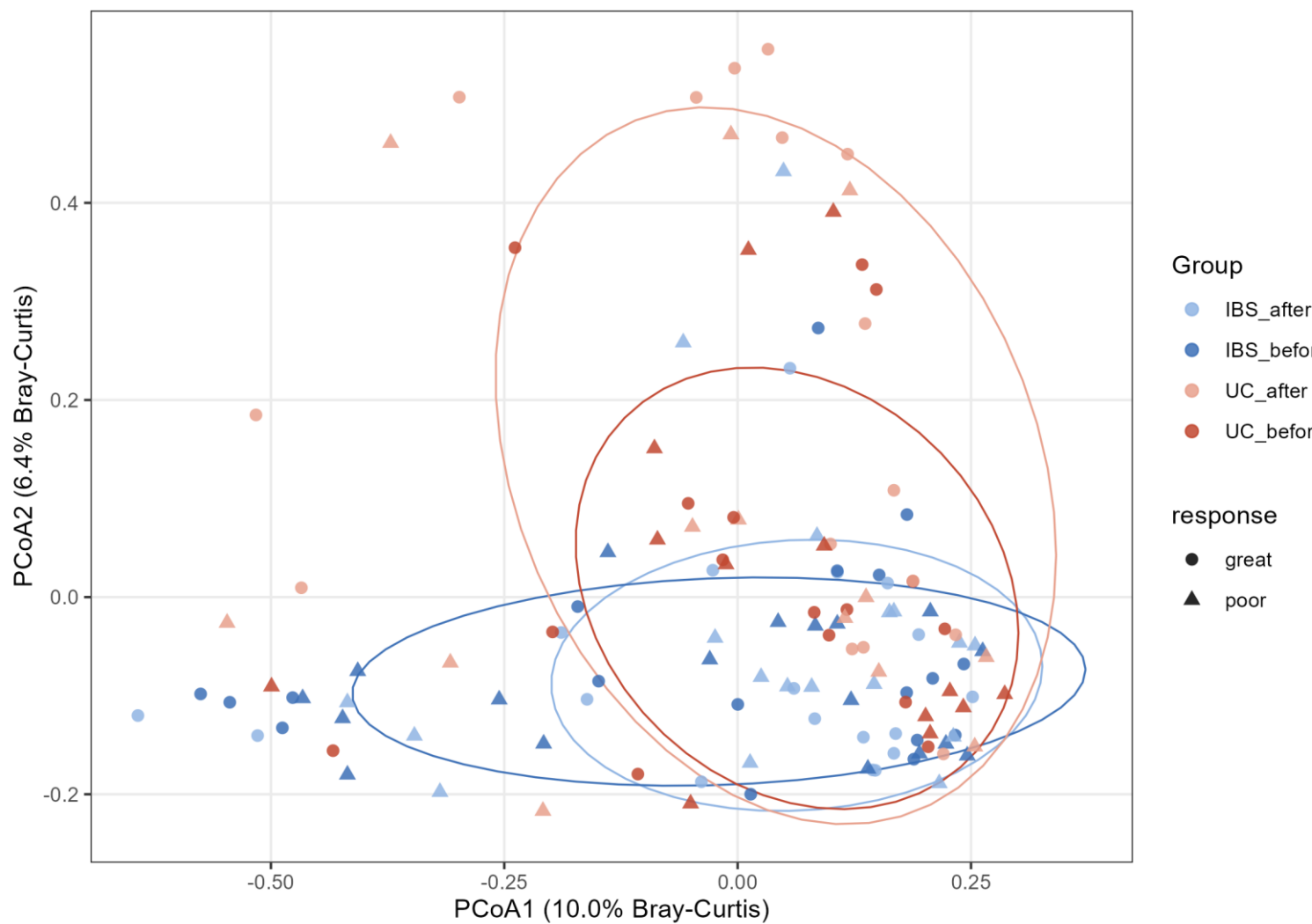
8. Figures (files in figures/)

Independent R analysis — alpha diversity with individual points, Bray–Curtis PCoA with axis variance and dispersion diagnostic:



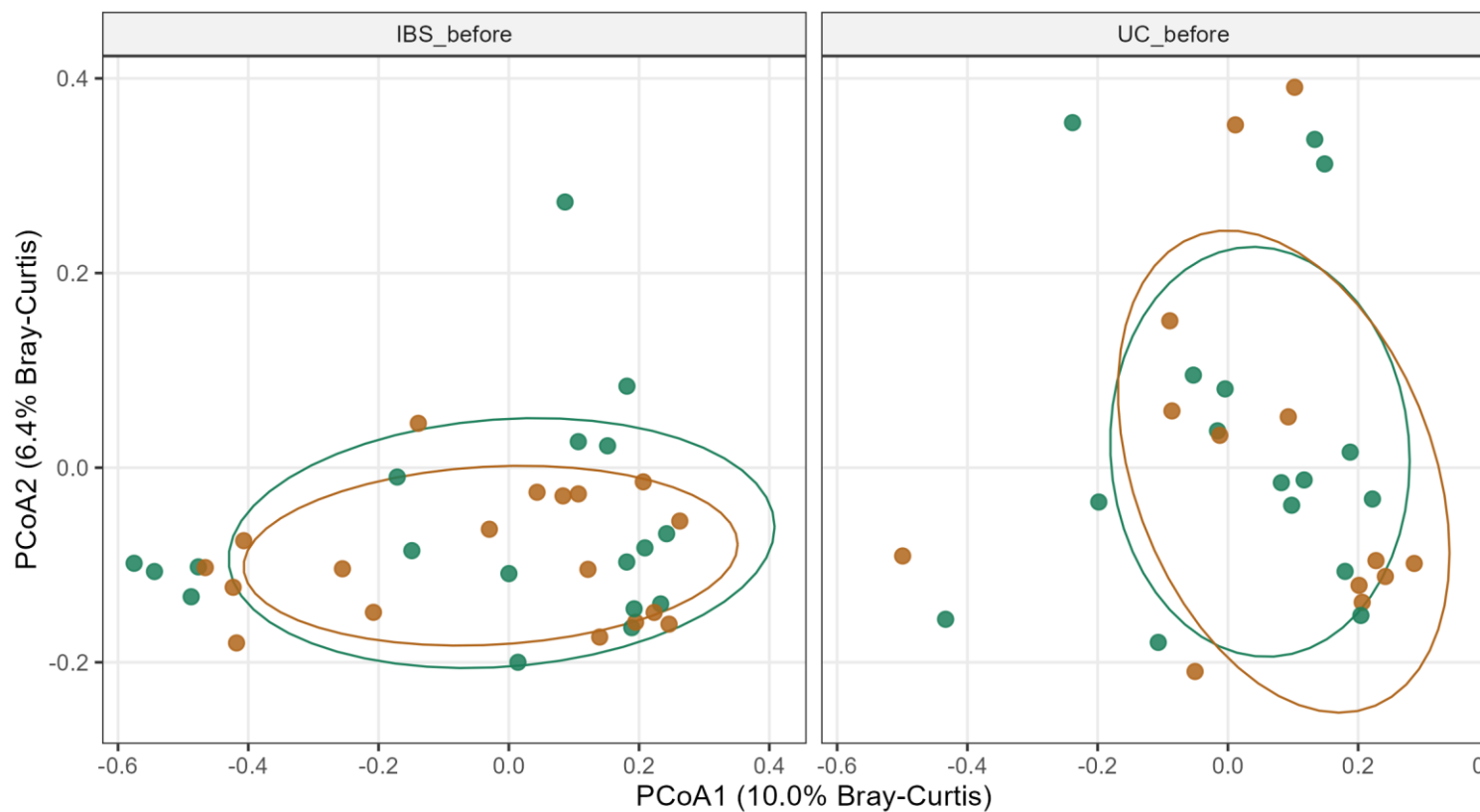
Bray-Curtis PCoA of 130 16S samples

PERMANOVA ~ Group: R2 = 0.047, p = 0.001; dispersion p = 0.449 (see tables/)



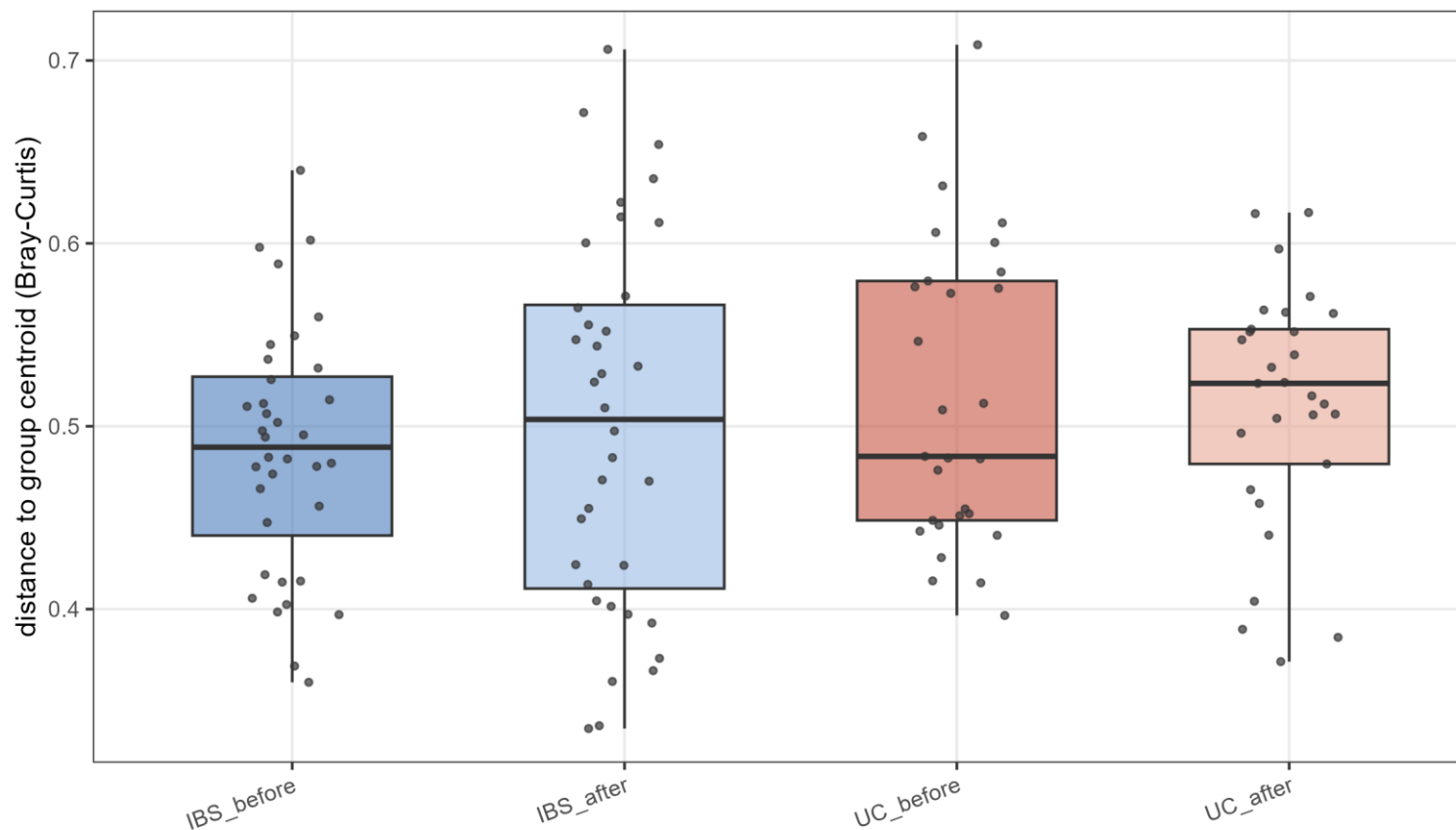
Baseline samples: response group (great vs poor) within each cohort

same ordination as above, restricted to pre-treatment samples

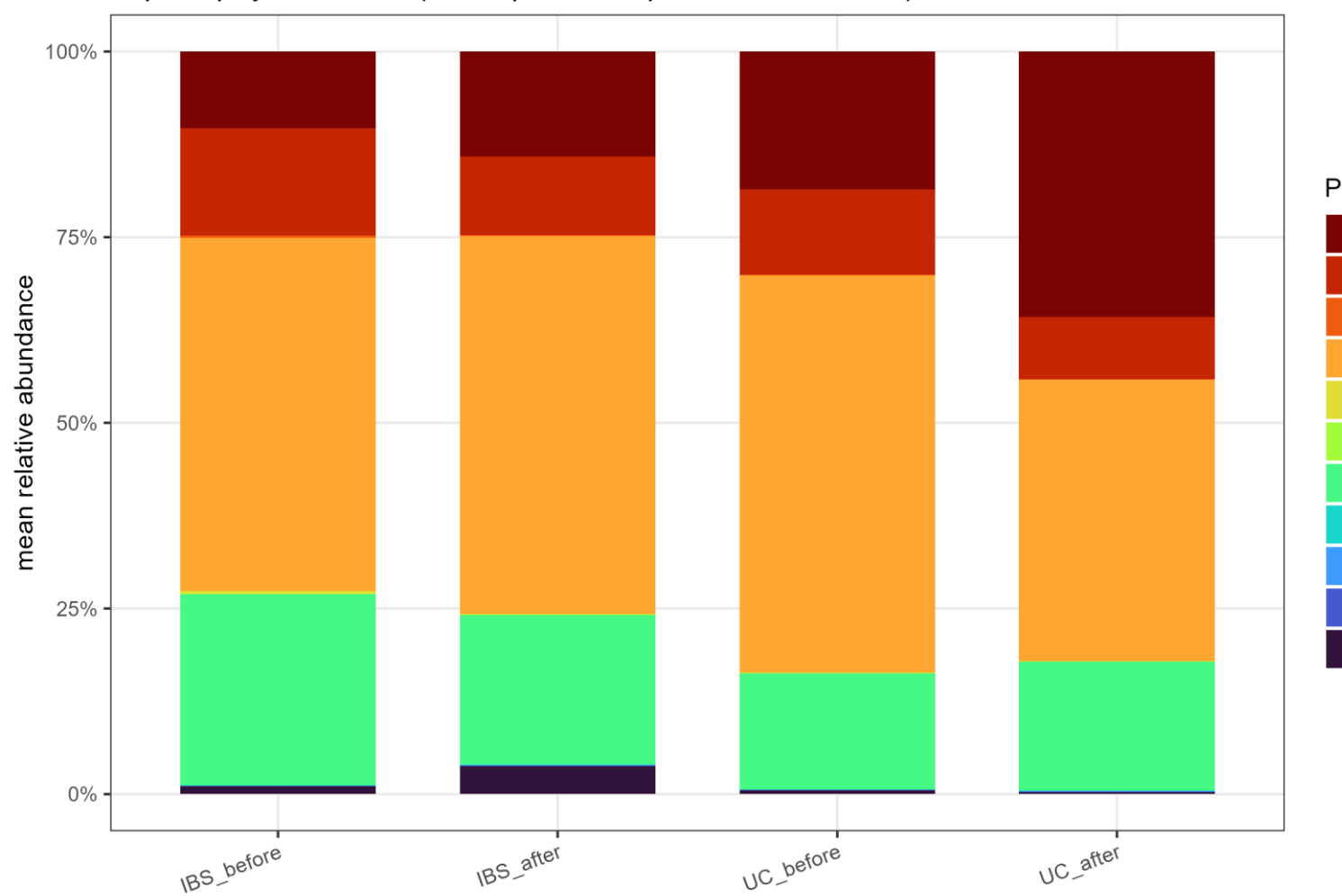


Within-group dispersion (betadisper)

PERMUTEST F = 0.86, p = 0.449 -> dispersion is checked before reading the PERMANOVA p-value

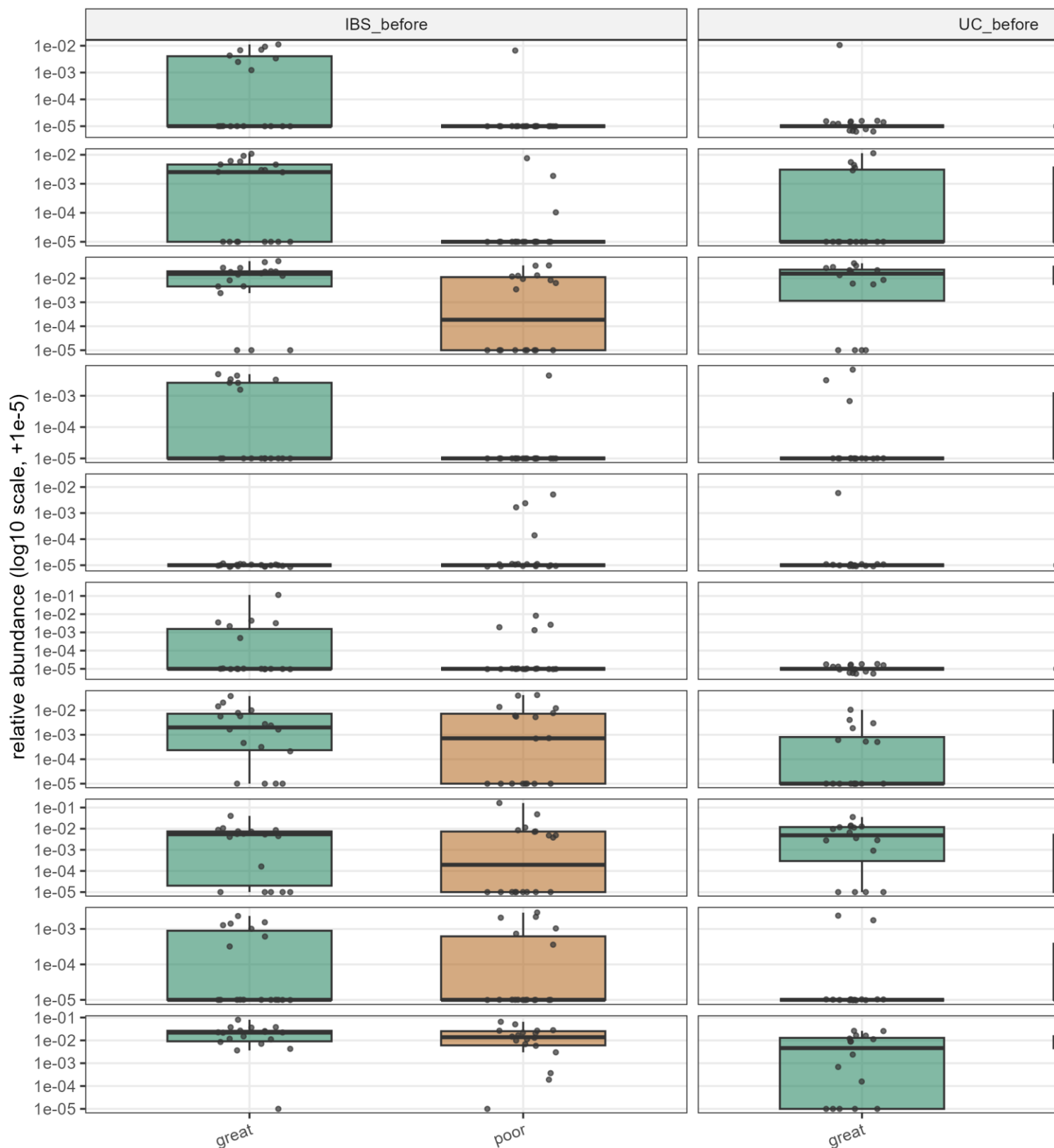


Top-10 phyla + Other (descriptive composition, not a test)



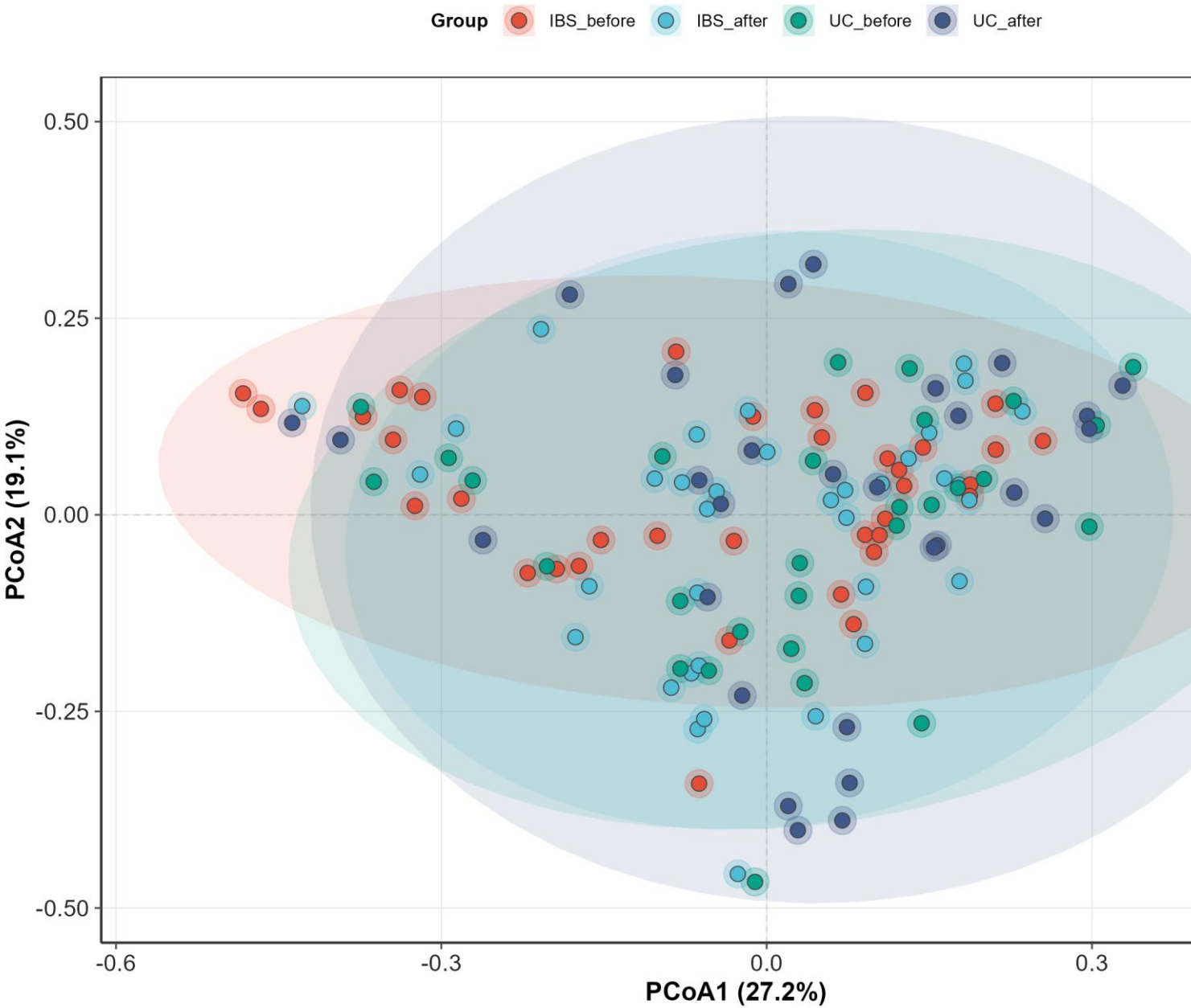
Top differential taxa at baseline (Wilcoxon, unadjusted $p < 0.05$)

exploratory; taxa with BH-FDR < 0.05 at baseline: 0 (IBS) and 0 (UC) - see tables/r_diff_taxa_wilcoxon.csv



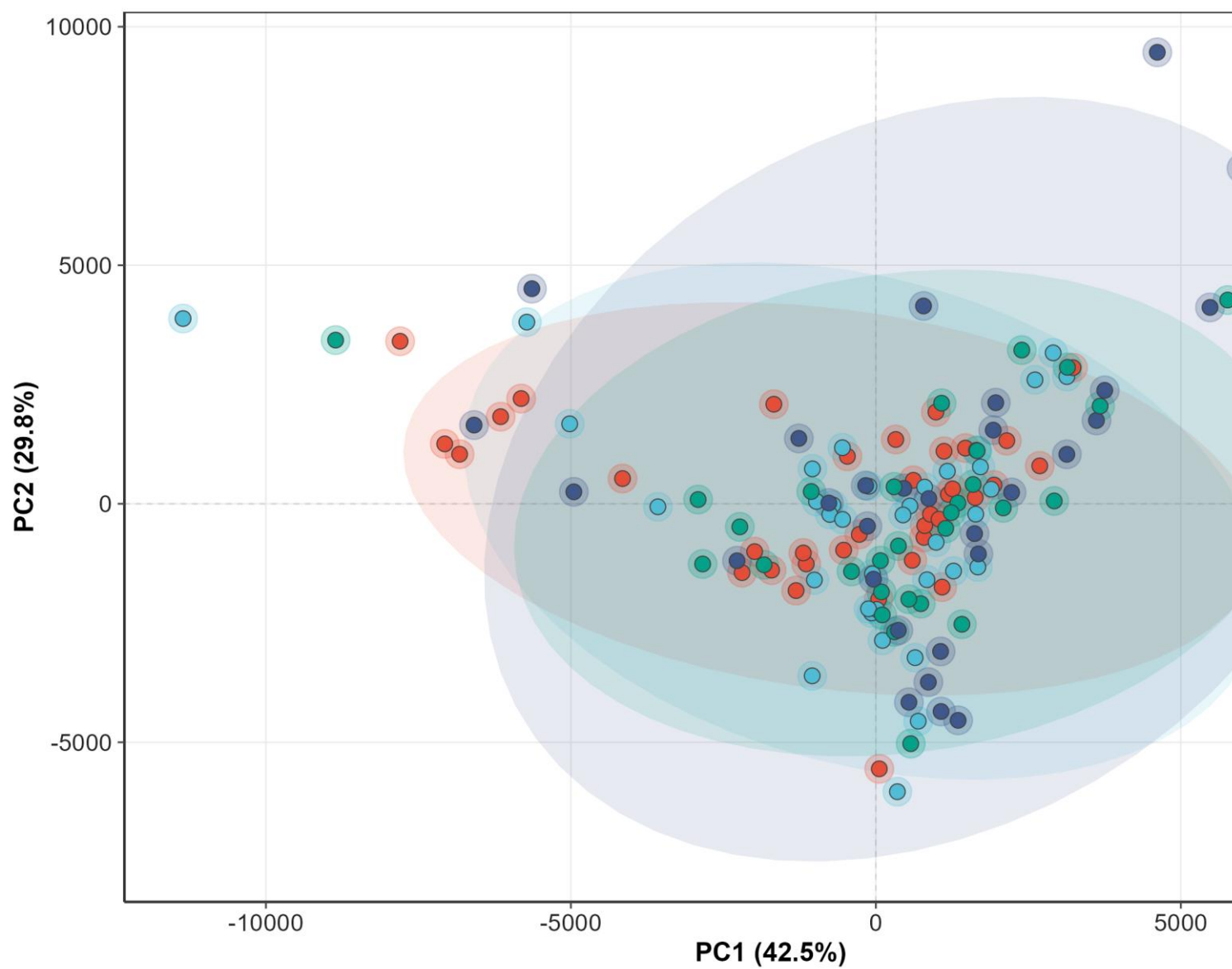
App-generated — the EMP Bray–Curtis PCoA (ordination="emp"), the app's default ordination (ordination="auto": Euclidean PCA of the raw assay, *not* the right data state), the top-40 heatmap, the sankey and the Spearman network:

PCoA (Bray–Curtis beta diversity)



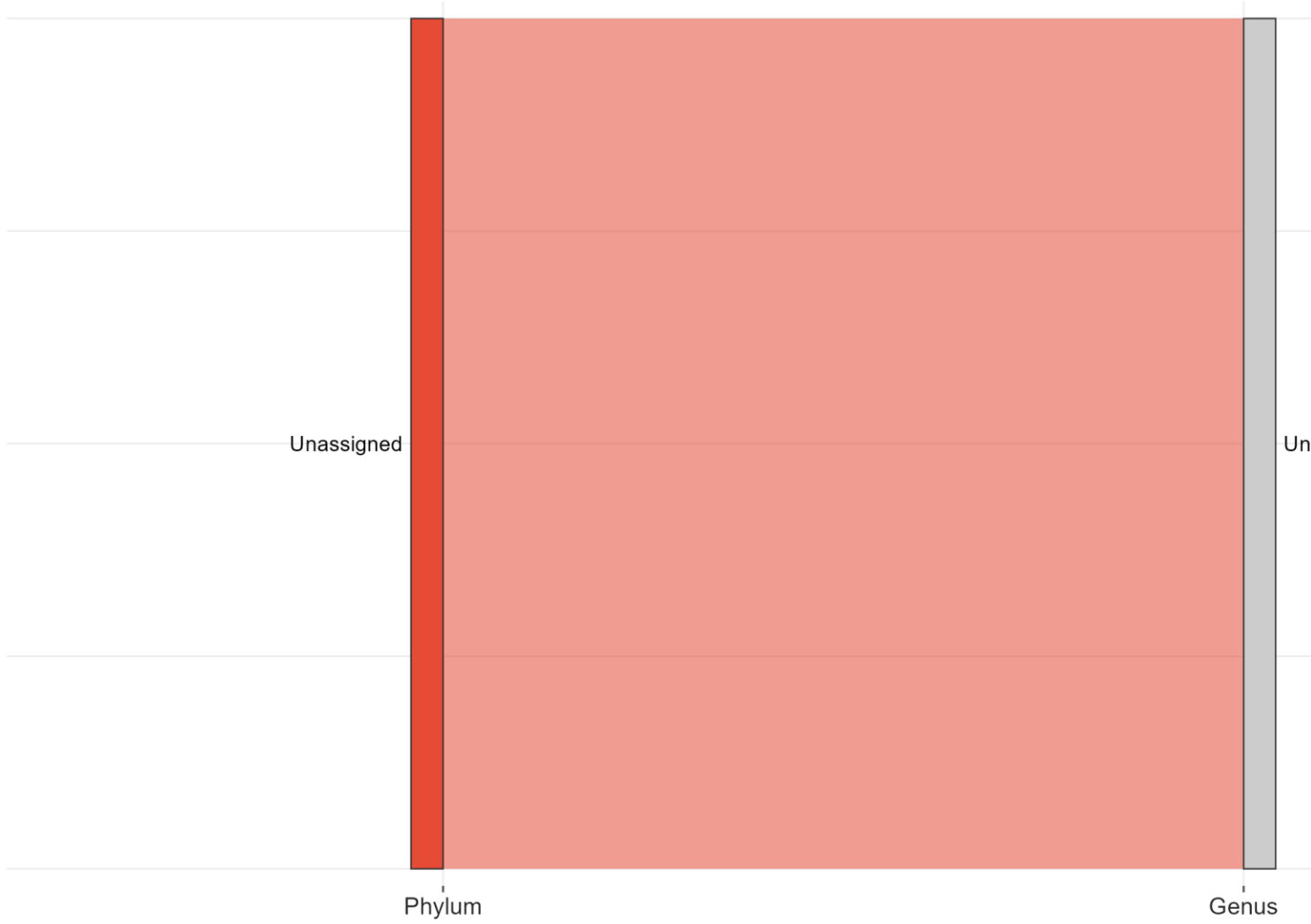
PCA (assay matrix, Euclidean)

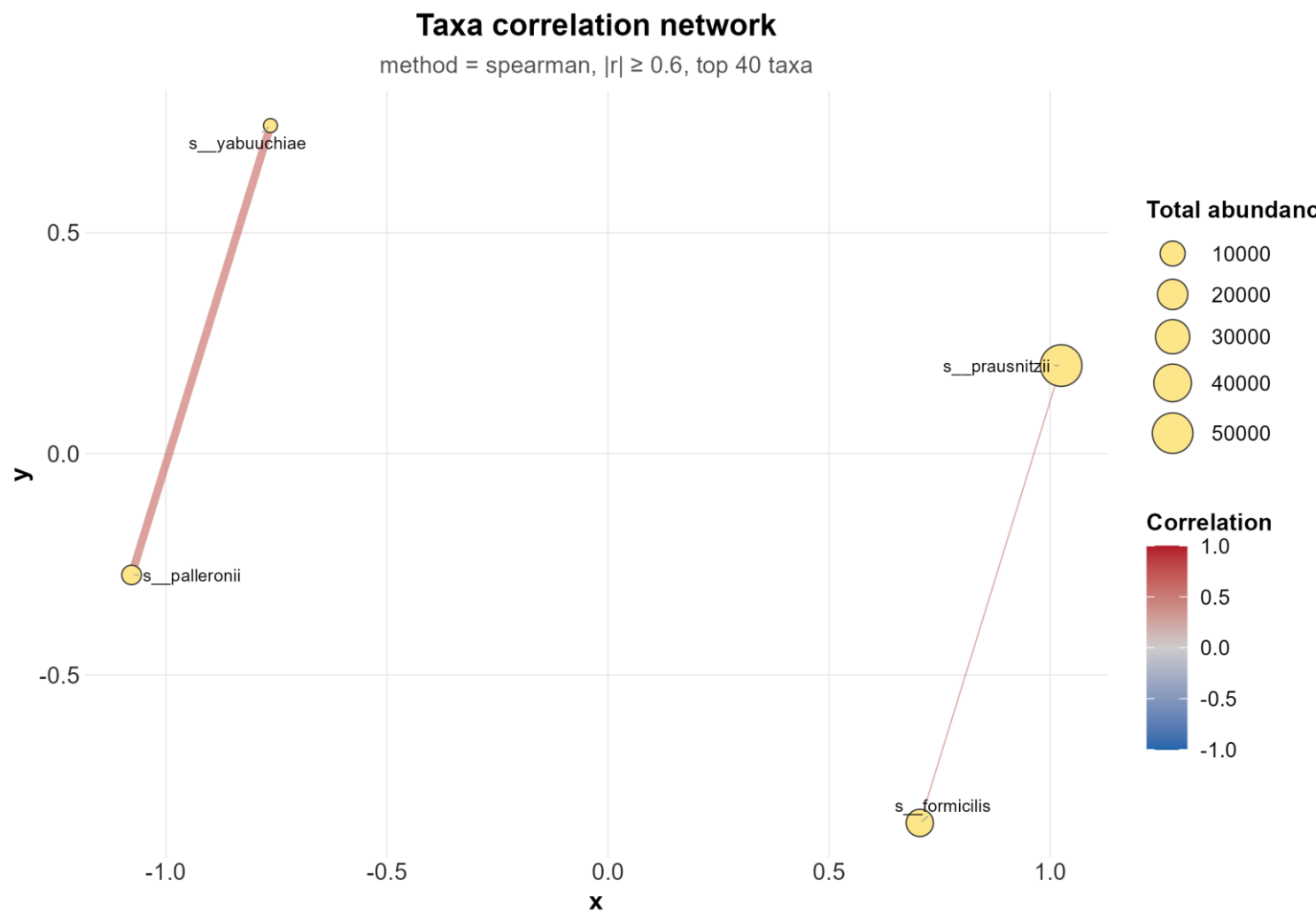
Group IBS_before IBS_after UC_before UC_after



Taxonomy flow: Phylum → Genus

Top 15 taxa, ribbon width $\propto$ abundance





Budgeted display versions of the app's own one-click plots (with axis titles and PDFs) are in `app_run_bundle/run2_analysis_130samples/plots/`.

9. Deliverable map (Week 6 reading-material rubric)

Rubric item	Where it is
R script, runs from a clean session, identity checks + comments	scripts/hw6_16s_verify.R (+ console/r_rerun_stdout.txt, console/hw6_16s_verify_console.txt)

Rubric item	Where it is
Summary table (depth, retained features, Shannon, PERMANOVA, dispersion)	tables/r_summary_table.csv (+ §4 table)
Alpha-diversity figure, points + group summary + test	figures/r_alpha_diversity.png, tests in tables/r_alpha_tests.csv
Bray–Curtis PCoA with interpretable axes	figures/r_pcoa_bray_curtis.png, figures/r_pcoa_baseline_response.png, figures/r_dispersion.png
Interpretation 100–150 words	§5 above (console/interpretation_wordcount.txt)
AI verification log + sessionInfo()	AI_verification_log.md, console/hw6_16s_verify_console.txt
EMP-web "complete analysis procedure" + submission	§2, §7, app_run_bundle/, app/api_responses/, app/api_call_log.md
Hypothesis + supporting parameters	§6

10. Reproduction

```
# 1) app analysis (see scripts/README_app_run.md for the full launch line)
cd C:/Users/郑烁瞳/EasyMultiProfiler-Web
R_LIBS_USER=D:/R/R-userlib API_HOST=127.0.0.1 API_PORT=8010 EMP_ALLOWED_ROOTS=C:/emp_hw6 \
  Rscript --vanilla webapp/backend/run_api.R
D:/python/python.exe homework1/scripts/prep_inputs.py      # 132 -> 130 samples (QC)
D:/python/python.exe homework1/scripts/stepA_official.py   # session, import, one-click run, bundle
D:/python/python.exe homework1/scripts/stepB_deep.py Genus # collapse, alpha, ordination, 10 contrasts
D:/python/python.exe homework1/scripts/stepC_appplots.py  # app figures
# 2) independent verification + figures + summary table
cd homework1/scripts
R_LIBS_USER=D:/R/R-userlib Rscript --vanilla hw6_16s_verify.R
```

Environment: R 4.4.3 (D:/R/R-4.4.3), libraries D:/R/R-userlib (vegan 2.7-3, ggplot2 4.0.3, edgeR 4.4.2, ragg 1.5.2), Python 3.11 (D:/python/python.exe), EasyMultiProfiler (Web) v9.0.4. Full sessionInfo() at the end of console/hw6_16s_verify_console.txt.